# Sagnik Singha

+91 9883358860 | sagnik.singha@outlook.com
github.com/cataclysmnik | linkedin.com/in/sagnik-singha-vit | sagniksingha.vercel.app

## Education

**Vellore Institute of Technology**
B.Tech. Computer Science & Engineering (Artificial Intelligence & Robotics)
Expected Graduation: 2028 | CGPA: **8.79/10.0**

## Experience

**UI/UX Design Intern – EaseLearnAI** (Jan 2026 – Present)
• Designed intuitive user interfaces and user experiences for educational web platforms.
• Created wireframes, prototypes, and reusable design systems in Figma.
• Collaborated with developers to implement responsive interfaces.
• Improved usability, accessibility, and visual consistency across products.

## Projects

**URSA – Unified Rocket Simulation Platform**
Technologies: **Python, PySide6, RocketPy**
• Unified RocketPy and other rocket simulation tools into a single desktop platform with an intuitive GUI for configuring, visualizing, and analyzing rocket simulations.

**Satellite Image Super Resolution**
Technologies: **Python, OpenCV, Machine Learning**
• Built a machine learning pipeline for infrared satellite image super-resolution and RGB image generation to improve interpretation of remote sensing imagery.

**Graphite – Open Source DAW**
Technologies: **Python**
• Developing a minimalist digital audio workstation for guitarists featuring Smart Comp, Backing Track Manager, and streamlined recording workflows.

**EDOSH – Educational Linux Shell**
Technologies: **C, Linux**
• Built an educational Linux shell implementing shell commands, process management, and interactive features for learning Unix concepts.

**Theta Weather**
Technologies: **React Native, Expo, WeatherAPI**
• Developed a cross-platform weather application with real-time forecasts, location-based weather updates, and a responsive interface.

## Technical Skills

**Languages:** Python, C, C++, Java, JavaScript, TypeScript, SQL, HTML, CSS
**Frameworks & Libraries:** React, Next.js, React Native, Expo, Node.js, Express.js, Tailwind CSS, PySide6, OpenCV, MediaPipe
**Tools & Technologies:** Git, Linux, Figma, Blender, Arduino, ESP32, Firebase

## Leadership

**Creative Lead – SEDS Antariksh**
Led the Design & Content team for student aerospace initiatives, managing branding, technical presentations, and outreach.

**Software Engineer – Team Ignition**
Developed software for student rocketry projects, contributing to simulation, embedded systems, and mission-support tools for rocket design, testing, and flight operations.

## Certification

Harvard CS50x: Introduction to Computer Science